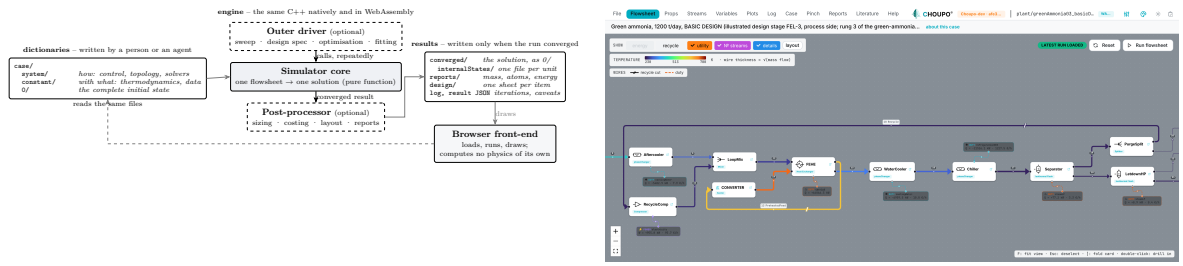
Graphical Abstract

Choupo: an open-source chemical process simulator for education and research

Vítor Geraldes, Pedro Mendes, Miguel Rodrigues



Every iteration visible · units on every value · data with source and licence · balances from the engine's own reports · unknown inputs refused by name

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Highlights

Choupo: an open-source chemical process simulator for education and research

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- An open process simulator that enforces six observables where inputs are read
- Convergence, physical consistency and acceptance are reported separately
- Regressed parameters must pass an independent study within a pre-declared band
- A fitted nanofiltration pore holds in rejection but not in transmission
- With identical rate laws, two simulators still differ by 0.6 % in conversion

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Choupo: an open-source chemical process simulator for education and research

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Abstract

Choupo is an open-source (GPL-3.0-or-later) sequential-modular process simulator written in C++17, for teaching and for membrane and electrolyte research. Its design goal is that six things are observable in plain-text files and the run log: the data a calculation used and their sources, its hypotheses, its equations, whether its parameters lie inside their evidence, its iterations, and its results with conservation balances. We argue that the contribution is not any one of these features, several of which exist in other open tools, but their enforcement where every input is read and their verification across a corpus of 494 cases. We describe the architecture, a curation rule that accepts a regressed parameter set only if it reproduces an independent study within a band declared before the fit, and why that band is conservative; and a verification method that separates numerical convergence, physical consistency and acceptance. The Cavett recycle problem converges in seven Newton steps and closes its first law to 0.002 kW of 396 MW exchanged, after a mixer defect the balance report had exposed was fixed. Nanofiltration is treated with its equations: a DSPM-DE pore fitted to measured chloride rejections predicts the held-out sulfates within one rejection point but not on the transmission scale, and a solution-diffusion-electromigration law reproduces the published trace-ion curves in the trace limit but not at the experiments' own concentrations. A comparison with an open-source simulator is split into a same-equations exercise, where a 0.6 % conversion difference remains, and a different-models exercise. No classroom evaluation has been made.

Keywords: process simulation, chemical engineering education, open-source software, thermophysical data curation, verification and validation, nanofiltration

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1. Introduction

The problem. A student of process simulation meets, in most exercises, converged numbers: a column specified and solved, a recycle that closed, stream tables to four decimals. Whether a missing interaction parameter was taken as zero, whether a vapour pressure was evaluated outside the range it was fitted on, whether a recycle converged or merely stopped, and whether the converged plant conserves energy, is information that a simulator holds but that the exercise does not make the student look at. The same questions matter to a researcher who inserts a new transport law into a plant model and must know which of the plant's numbers the new law moved. Choupo was written so that those questions

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have answers a reader can find in files. Whether this improves learning is a goal, not a result: no classroom evaluation exists, and we make no claim of one.

Six observables. We call a simulator a *glass box* when a user can observe, in plain-text files and in the run log, and without changing a default:

(i) the data a calculation used, and the source of each value;

(ii) the hypotheses the calculation rests on;

(iii) the equations it solves;

(iv) the parameters it took, and whether they lie inside their evidence;

(v) the numerical method and its iterations;

(vi) the results, with their conservation balances.

State of the art, property by property. Several open-source tools provide some of these observables, and some provide them better than Choupo. Table 1 compares four classes of tool on each one. The equation-oriented framework IDAES [1] writes every model as algebraic constraints that a user can list and inspect programmatically, so (iii) is stronger there than in any compiled simulator; its balances are constraints of the model, satisfied to the solver's tolerance. The property library CoolProp [2] attaches the literature reference of each fluid's equation of state to the fluid, which is (i) for pure-fluid properties; Cantera [3] reads mechanisms from input files whose species carry declared temperature ranges. DWSIM [4], an open-source sequential-modular simulator with a graphical editor, offers the breadth of unit operations and property packages that Choupo does not seek. What none of these tools does, as far as we could establish from their documentation, is to *enforce* the six observables where the input is read: in Choupo a dimensional value without a unit, a model word the engine does not know, an over-specified stream or an unauthorised approximation is refused before anything is computed; a value from a record outside its declared validity window is announced each time it is used and repeated at the end of the run; and the three conservation balances are computed after the solve from the streams, by a reporting layer that does not reuse the units' arithmetic, on every steady run. The contribution we claim is that integration and its verification across a corpus (Section 3); none of the individual features is new.

Agents. An AI agent can now write a Choupo case from a sentence, fit a model to an archive's data and run the result. That makes a run cheap and an understood run no cheaper: the formulation, the hypotheses and the reading of the result are still the student's. The six observables are what makes an agent's work checkable by a person, and the repository ships a short reference an assistant is given before it writes a case (`AGENTS.md`, `docs/ai/`). We present this as a design goal; we have measured nothing about it.

Contributions. (1) An architecture in which the six observables are enforced at the point where inputs are read (Section 2). (2) A verification method that separates numerical convergence, physical consistency and acceptance, with its residuals, tolerances and coverage stated (Section 3). (3) A curation rule for regressed parameters, argued statistically, with an example in which the candidate fails (Section 4). (4) Results on four problems: a flash and a heater checked by hand, the Cavett recycle problem, nanofiltration of multi-ionic solutions against measurements, and a two-part comparison with an open-source simulator (Section 5). Section 6 gives the limitations.

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Table 1: How each of the six observables is reached in four classes of open-source tool, as read from each tool's documentation. "Not established" marks a cell the authors could not settle from the documentation; it is not a claim that the feature is absent. The last column is what Choupo does by default.

	IDAES (equation-oriented framework)	**DWSIM** (flowsheet simulator, GUI)	**CoolProp, Cantera** (property and kinetics libraries)	**Choupo**
(i) data and sources	parameters in the property-package source; citation by convention	compound databases; per-value source not established	CoolProp: references per fluid equation of state; Cantera: optional notes in input files	primary source and licence per value in each record; records without a citation are listed by name
(ii) hypotheses	model configuration in the user's script	property package and unit options chosen in the GUI	the library's model	declared in the case; every default the run uses is announced and replayed at the end
(iii) equations	Pyomo constraints, inspectable programmatically	open source code	open source code and documentation	open source code and a theory guide; the log names the model each unit used
(iv) parameters vs evidence	variable bounds; validity ranges not established	not established	CoolProp: input limits of each equation of state; Cantera: temperature ranges of species data	every declared validity window checked at evaluation; extrapolation announced
(v) method and iterations	the nonlinear solver's log	not established	not applicable	every Newton iteration of every unit and recycle at the default verbosity
(vi) balances	balances are model constraints, met to solver tolerance	not established	not applicable	mass, element and energy balances recomputed from the streams after the solve, on every steady run

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2. The software

2.1. Scope

Table 2 says what Choupo models; every count is read from a file the repository regenerates from its own tree. Choupo is written in C++17 with no external library and builds with `g++` and GNU make; substantial parts of the code were written by AI coding agents under the authors' written briefs and reviewed before integration (see the declaration at the end of the paper), and the code a student reads is ordinary C++ with no generated sources or registration macros.

2.2. The case format

A case is a directory. `system/` says *how* (control, flowsheet topology, solver options, an optional outer driver and post-processing); `constant/` says *with what* (the thermodynamic system, reactions, case-local data); `0/` holds the complete initial state, one file per stream, and a missing or orphan stream file is a fatal error. Every dimensional scalar carries a named unit (`P 1 bar;`) or a dimension set and an SI value; the parser converts to SI and refuses an unknown unit or a dimension mismatch by name. Listing 1 is the whole topology and feed of the first tutorial. The steady solution is written to `converged/` in the layout of `0/`, only when the run converged, with each unit's interior (a column's stage profile, a reactor's axial profile) beside the streams, so that a converged directory renamed `0/` restarts the case. Every tutorial is *sealed*: the property records it reads are copied into its own `constant/` under a manifest with a checksum per record, and the installation's catalogue is forbidden to it, so that it answers the same on any machine.

Table 2: What Choupo models (development line; counts from `generated/releaseInventory.json` and `generated/methodTools.json`).

Binaries (5)	steady state (recycles by Newton on declared tear streams, or Wegstein); batch campaigns driven by a recipe; a time-integrated flowsheet without control loops; the same with a control layer; property evaluation, scans and parameter fitting.
Unit operations (52)	flashes (incl. three-phase); reactors (conversion, equilibrium, Gibbs, CSTR, PFR incl. cooled multitubular, batch); distillation (Wang–Henke, simultaneous MESH, shortcut), absorbers, strippers, extractors; heat exchangers, evaporators; crystallisers; dryers; spiral-wound NF/RO modules, electrodialysis stack and batch rig, ion exchange; adsorption; solids handling; compressors, turbines, pumps, mixers, splitters, valves, pipes, tanks.
Thermodynamics	ideal, NRTL, Wilson, UNIQUAC, UNIFAC, COSMO-SAC (2002); ideal gas, SRK, Peng–Robinson, PC-SAFT with association; single-salt Pitzer and electrolyte NRTL; Davies, Pitzer–HMW and Edwards–Pitzer speciation; IAPWS-IF97.
Catalogue	604 components, 53 aqueous species, 276 mixture parameter records, 4 membranes, 3 adsorbents, 10 utilities.
Plant scale	sectored plants with a thermodynamic model per sector; parameter sweeps, design specifications, optimisation, parameter estimation; sizing, bare-module costing, cash flow, plant layout.
Corpus	494 runnable cases, 477 with a recorded reference answer; 65 interactive lessons in the browser.

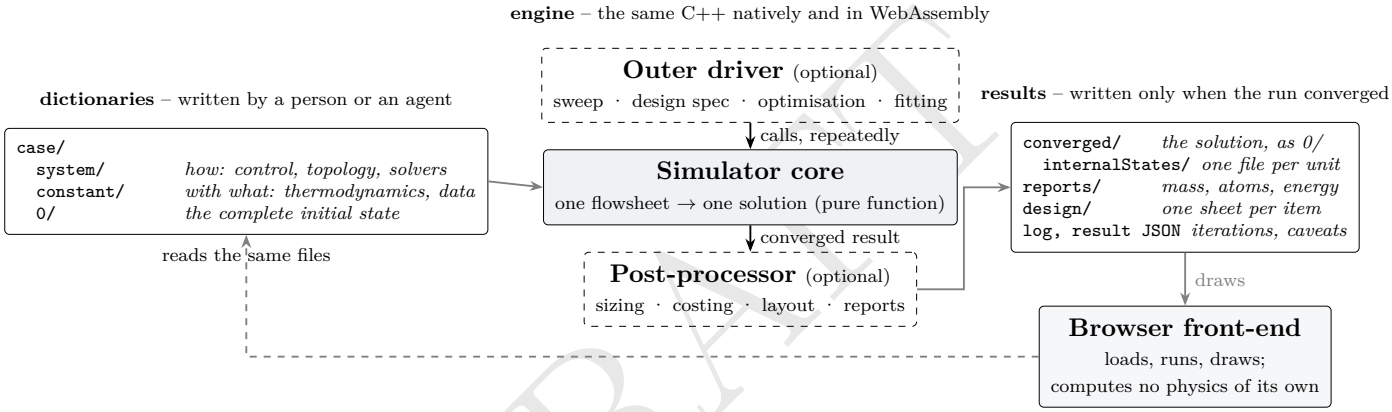


Figure 1: Dictionaries, engine, results and front-end. The three layers of the engine; dashed layers are optional. The browser front-end runs the same compiled engine and draws what it published.

Listing 1: The first tutorial's topology, `system/flowsheetDict`, and its feed, `0/feed` (case files, GPL-3.0-or-later; banners omitted).

```
units
(
    {
        name        flash01;
        type        isothermalFlash;
        model       rachfordRice;
        in          feed;
        outputs     ( liquid  vapor );
        operation   { T 370.0 K;  P 1.0 bar; }
    }
);
------------------------------------------- 0/feed
componentMolarFlows { benzene 40 kmol/h;  toluene 60 kmol/h; }
T               370 K;
P               100000 Pa;
```

2.3. Architecture

The engine has three layers (Figure 1). The simulator core maps one flowsheet to one solution as a pure function of its inputs. An optional outer driver calls that function repeatedly (sweeps, design specifications, optimisation, parameter estimation), and an

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optional post-processor chain adds sizing, costing and layout. Numerical strategies coexist in each binary and are chosen in the case: Newton and Wegstein [5] on declared tear streams, Nelder–Mead [6], Levenberg–Marquardt [7], Michelsen's stability test [8], explicit and Rosenbrock integrators. Every extensible base class (unit operations, thermodynamic models, drivers, sizers) has a factory, and every built-in type is registered by one line in a function `main()` calls explicitly, so the origin of a model is one readable function. A user may add a unit operation or a rate law by the same mechanism, compiled into a per-case binary on a local installation, or write a rate law as a dimension-checked formula in the case file. The browser front-end runs the same C++ compiled to WebAssembly; it loads the dictionaries, runs the engine and draws what the engine published, and computes no equilibrium, balance or size itself.

OpenFOAM. The case organisation (`system/`, `constant/`, `0/`, keyword–value dictionaries, time directories) follows OpenFOAM®[1] [9]; the main departures are one binary per problem class rather than per algorithm, explicit registration instead of registration macros, and mandatory units on every dimensional value. **Choupo is an independent project: it is not part of, affiliated with, endorsed by or sponsored by OpenCFD Ltd or the OpenFOAM Foundation, and no OpenFOAM source code was adapted or reused in it.** *This offering is not approved or endorsed by OpenCFD Limited, producer and distributor of the OpenFOAM software via www.openfoam.com, and owner of the OPENFOAM® and OpenCFD® trade marks.*

3. Verification and validation: the method

3.1. Three questions kept apart

A run answers three different questions, and Choupo reports each separately.

Numerical convergence – did the iteration meet its declared criterion? Each solver declares a tolerance, a relative tolerance and an iteration limit; convergence is decided on a normalised residual (one home, `solver/Convergence.H`) and the run prints the raw and normalised residuals and the criterion that decided. `converged/` is written only when the steady solve converged, and means exactly this and nothing more.

Physical consistency – does the converged answer conserve mass, atoms and energy? Three balances are computed after the solve, from the stream states, by the reporting layer (Section 3.2), and printed whether or not the case asked for them. A converged run may be inconsistent, and the balance report is how a user sees it: the earlier draft of this paper showed a converged Cavett plant whose first law missed by 88 % (Section 5.2).

Acceptance – is the answer right for its purpose? That needs an independent reference: a measurement, a published solution, a hand calculation. A regression pass does not answer it (below).

[1] OPENFOAM® is a registered trade mark of OpenCFD Limited, producer and distributor of the OpenFOAM software via www.openfoam.com.

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Table 3: Coverage of the two conservation checks over the 286 steady cases of the corpus (494 cases in all; the other 208 are batch, dynamic and property cases, whose balances are held by other checks or not at all). Each count is the check's own claim line on the tree of this revision.

	mass	**energy**
close within the threshold (energy: in %, or in kW where no % exists)	261	113 + 71
listed by name as not closing (residual pinned)	–	5 + 7
balance refused for want of an enthalpy datum on a boundary stream	–	70
no material boundary (closed cycles)	4	–
report did not run under an outer driver (announced)	1	1
no report of that kind printed	13	13
exit non-zero (held by the suite's exit-code check)	7	7
total	286	287[a]

[a]The "listed" row of the energy column is the length of the check's two lists (5 + 7), so one listed plant is also counted in another category of this pass; the column sums to 287 for 286 cases.

3.2. The balances, their normalisation and their tolerances

For a plant with boundary inlets $\mathcal{I}$ and outlets $\mathcal{O}$ (streams internal to the flowsheet excluded) the reports compute

$$\text{mass closure} = 100\,\frac{\sum_{s\in\mathcal{O}}\dot{m}_s}{\sum_{s\in\mathcal{I}}\dot{m}_s}\ \%, \qquad \text{element closure}_e = 100\,\frac{\sum_{s\in\mathcal{O}}\dot{n}_{e,s}}{\sum_{s\in\mathcal{I}}\dot{n}_{e,s}}\ \%, \tag{1}$$

$$r_E = \Big(\sum_{s\in\mathcal{I}}\dot{H}_s + Q + W\Big) - \sum_{s\in\mathcal{O}}\dot{H}_s, \tag{2}$$

$$\varepsilon_E = 100\,\frac{r_E}{\max\Big(\sum_u |Q_u| + \sum_u |W_u|,\ |Q|\Big)}\ \%, \tag{3}$$

where $\dot{H}_s$ is each stream's enthalpy on the elements-and-formation datum, priced by the thermodynamic model of the unit it leaves after resolving its phase split, Q and W are the heat and shaft work crossing the boundary, and the denominator is the energy the plant *exchanges*. Dividing by the enthalpy throughput instead, which is dominated by the formation datum passing through unchanged, made a real gap look small; dividing by the terms the equation balances is the same rule the solver uses for its residuals. When a plant declares no duty and no work, the denominator vanishes, and the report prints the residual in kW with the statement that no percentage exists. The same report prints each unit's own residual, so a plant-level gap is localised.

Two automated checks read these reports on every steady case of the corpus and never recompute them: mass closure within 100 ± 0.5 %, and $|\varepsilon_E| \le 1$ % or, where no percentage exists, $|r_E| \le 1$ kW. These numbers are detection thresholds, not accuracy claims, and they are conventions: a plant with recycles converges its tear streams to a declared relative tolerance (typically 10^{-6}), so it closes to a residual and not to zero, and the thresholds are the loosest values that are still a physical statement while sitting three or more orders of magnitude above what a converged tear explains. A plant losing 0.4 % of its mass passes; the defect the mass check was built for created 13 %. Plants known not to close are listed by name with their measured residual, and a listed residual that grows by more than 0.2 percentage points, changes sign, or closes without the list being updated fails the check, so that the list keeps describing the engine as it is. Table 3 reconciles the coverage.

Independent checks of the reports. The balance reports are themselves code, so a few were checked by hand. The opening flash (Listing 1) is fed at its own temperature and pressure

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and exchanges no heat, so its zero energy residual tests nothing; its equilibrium can be tested instead: an independent bisection of the Rachford–Rice equation on the case's own Antoine constants gives $V/F = 0.30398357$, the engine's fourth Newton iterate. A heater is the smallest non-trivial energy test: 100 kW into 100 kmol/h of equimolar ethanol and water with constant liquid heat capacities of 112.3 and 75.5 $\mathrm{J\,mol^{-1}K^{-1}}$ (ideal liquid) gives $T_{\mathrm{out}} = 300 + 3600/93.9 = 338.33866$ K by hand; the engine publishes 338.3386581 K and a first-law residual of 0.000000 kW over 100 kW exchanged. Mass and element closures are checked the same way on the flash, where $z_i = (1 - V)x_i + Vy_i$ holds to the six printed digits. These are verification of the arithmetic, not validation of the models.

3.3. Regression is not validation

Of the 494 cases, 477 carry a reference answer that an earlier Choupo run recorded; each suite run compares every case with it and screens every case for NaN and infinity. A pass proves that an answer has not moved, not that it is right; a re-recorded reference is therefore listed row by row, each with the reason it moved, before anything is re-recorded. A regression check protects the behaviour and the conditions it exercises, and no more: a case the suite does not run, or a condition no case reaches, is outside it. Following the distinction of Oberkampf and Roy [10] and NASA-STD-7009 [11], validation is a separate, named subset of cases anchored to independent published results – the Williams–Otto reactor [12], the IAPWS-IF97 verification tables [13], the Hamer–Wu activity coefficients [14], a measured calcite pH [15] and others; a case whose anchor is its own training data is labelled fit-consistency and not counted. The membrane and electrodialysis comparisons of Section 5.3 are with measurements but are not yet in the named subset. The subset holds no rigorous column, recycle plant or liquid–liquid flowsheet anchored to a measurement.

3.4. Refusal by name

An input the engine cannot honour is refused before anything is computed, with the accepted alternatives or the remedy. Three classes are refused this way: a word the engine dispatches on and does not know (one function lists the accepted words and the closest match); an input that over-specifies the problem, such as a mixture stream with a temperature, a pressure and a vapour fraction (Listing 3); and an approximation the case did not authorise, such as a missing binary pair silently replaced by ideal behaviour, in the activity models where this decision is wired. Each class is held by an automated check that builds the offending input and requires the refusal and its wording. When a class of error has reached a user before such a check existed, the check was written afterwards and verified by reintroducing the defect; Appendix Appendix A lists the checks behind the examples of this paper. A check protects the conditions it exercises; it does not make a class of error impossible.

4. Thermophysical data curation

4.1. A source and a licence for every value

Each record in the curated catalogue cites the primary source of each value and how it was obtained (measured, regressed, estimated); a record not yet reviewed, or read from the user's private tier, is announced each time a run reads it. Licence classes are decided before a value enters the tree: a dataset with no licence statement is treated as no-grant and kept out, as is a NonCommercial grant. Four records whose source has an encumbered licence are not yet replaced; they are listed by name by the licence check rather than kept

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silently, so the catalogue's claim is "a source and a licence for every value, except four listed records".

4.2. The acceptance rule

A regressed parameter set (for example the two parameters of an NRTL pair) is fitted on one study, the *fit set*, and scored on a study from another laboratory, the *held-out set*, by the mean absolute deviation of the predicted bubble temperatures, $\mathrm{AAD} = n^{-1}\sum_i |T_i^{\mathrm{model}} - T_i^{\mathrm{exp}}|$. The set is accepted when

$$\mathrm{AAD} \;\le\; b \;=\; \max_i U_{95,i} \;+\; \max\left(|\delta_1|,\, |\delta_2|\right), \tag{4}$$

with $U_{95,i}$ the expanded uncertainty (95 % coverage) the archive states for point i of the held-out set, and δ_1, δ_2 the error of the catalogue's vapour-pressure model at the two pure-component ends, measured with no pair involved. The band is declared, in the case file, before the fit is run, which prevents choosing it after seeing the residuals; that alone does not make it adequate, so we state its basis.

If the model were exact and the measurements carried only random error with standard uncertainty $u_i = U_{95,i}/2$ (coverage factor 2, Joint Committee for Guides in Metrology [16]), the expected AAD would be $\sqrt{2/\pi}\,\bar{u} \approx 0.4\,\bar{U}_{95}$, well below the first term of b. A pure-component vapour-pressure error enters a nearly ideal bubble temperature approximately as $x_1\delta_1 + x_2\delta_2$, whose magnitude never exceeds $\max(|\delta_1|, |\delta_2|)$. The rule therefore bounds noise and a systematic endpoint error by worst-case *linear* addition; it is conservative for a correct model and designed not to reject one. Its limitations are stated rather than hidden: the archive's combined expanded uncertainties are taken as given, and whether each source article included the propagated composition and pressure uncertainties in them was not checked; systematic errors other than the endpoints' (in the composition analysis, for instance) are not modelled; and a threshold on a mean cannot see a set that fails at a few points. For this reason the verdict is also reported against a χ^2 statistic, $\chi^2 = \sum_i (r_i/u_i)^2$ with n degrees of freedom, which uses every point's own uncertainty.

4.3. A worked example: a candidate that fails

The acetone + methanol pair is the azeotrope a student meets in two plants of the corpus, and the one of the regressed pairs that failed. An agent located two isobaric studies in the NIST/TRC ThermoML archive [17], one at 101.3 kPa [18] and one at 100 kPa [19], from two laboratories with no author in common, extracted each with its per-point uncertainty, set the pure ends aside, and wrote a case declaring the larger study as the fit set, the other as held out, two NRTL parameters (α fixed at 0.30), uncertainty weighting, and the band of Eq. (4): 0.155 K (0.047 % of the held-out mean temperature). The engine records the datasets with DOI, archive file, checksum and review status, freezes a fingerprint of the partition before the regression, fits by Levenberg–Marquardt from two starts, and computes the verdict; it also runs bubble-point scans with ideal behaviour, UNIFAC [20] and the fitted NRTL [21] at each study's pressure (Figure 2).

The fitted pair reproduces its own study to 0.10 K and misses the held-out study by 0.316 K, twice the band: the verdict is `notValidated` and the pair does not enter the catalogue. The verdict does not depend on the details of the rule: against the per-point uncertainties the held-out residuals give $\chi^2 = 1935$ on 30 points (95 % quantile 43.8), and noise alone would give an AAD of about 0.037 K. UNIFAC reproduces the held-out study to 0.041 K ($\chi^2 = 31$, consistent with the stated uncertainties) and misses the fit study by

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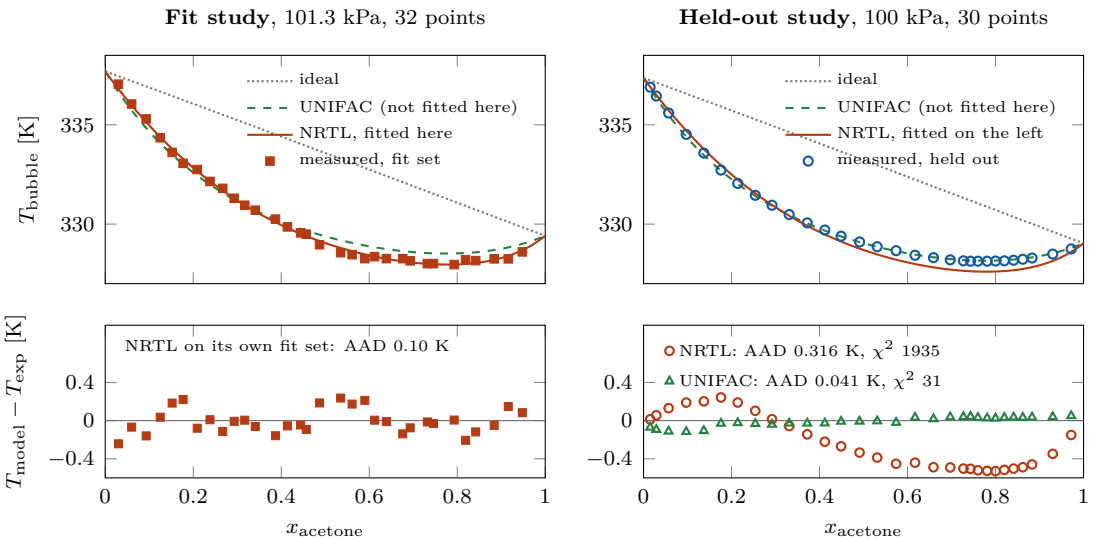


Figure 2: The acetone + methanol regression, drawn from the run's files. Top: bubble-point curves at each study's pressure, with the fit study (left) and the held-out study (right). Bottom: residuals per point. The acceptance criterion applies to the *mean* absolute residual (AAD), not to individual points: the band declared before the fit is $b = 0.155$ K; the held-out points carry stated expanded uncertainties of 0.09–0.11 K each. χ^2 is $\sum(r_i/u_i)^2$ with $u_i = U_{95,i}/2$ over the 30 held-out points; its 95 % quantile is 43.8. The fitted pair reproduces its own study and fails the other (verdict `notValidated`); UNIFAC does the mirror image.

0.34 K. UNIFAC was not fitted to either study *in this work*; its group parameters were regressed by their authors to a large VLE database that may include data for this binary, so its agreement is not evidence of independence. The two studies disagree with each other by about 0.3 K where each states 0.1–0.35 K, so at least one carries an error its uncertainty does not describe; which one is a question for a curator who reads both articles, and neither dataset has yet been read back against its article (the archive transcription was used; every run says so). For a column at these pressures both models place the azeotrope near $x_{\mathrm{acetone}} \approx 0.78$ at 100 kPa; we do not claim that a column design is insensitive to the choice, which depends on the equilibrium along the whole column and on how close the specification is to the azeotrope. A second candidate fails the other way: an ethanol + cyclohexane pair passed its held-out band and was not promoted, because it predicts two liquids across its validity span where both studies report one – a bubble-temperature test solves for one liquid by construction and cannot see a second, so acceptance also requires a stability scan.

5. Results

Every number in this section was read from a run of a named case on the tree of this revision; supplement S1 lists the commands.

5.1. A flash and a refusal

Listing 2 is an excerpt of the run log of the flash of Listing 1. It prints the model, the vapour pressures and K-values, each Newton iteration of the Rachford–Rice equation, that benzene's vapour pressure was evaluated outside the range of its Antoine correlation, and at the end every assumption the run raised. A stream of known component flows is fixed by two intensive variables; the same feed given a vapour fraction as well is refused before any computation (Listing 3).

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Listing 2: Excerpt of the run log of the flash (`verbosity 3`); elided lines marked `[...]`.

```
>>>  Unit [0]:  flash01   (type = isothermalFlash)
>>>  thermo: inherited (global package)
  [flash] model rachfordRice  (declared)
[psat] component 'benzene': Antoine evaluated at T = 370 K, OUTSIDE its declared Trange (287.7 354.07).
K-values  (γ = ideal,  φ = idealGas,  evaluated at x = z initially):
  benzene       Psat = 1.6514 bar    K = 1.6514
  toluene       Psat = 0.6735 bar    K = 0.6735
  [outer 0] Rachford-Rice Newton (inner):
    it          V           g(V)         dg/dV         ΔV
    0     0.50000000   -3.75806e-02   -1.87920e-01   -1.99981e-01
    1     0.30001863   7.81978e-04    -1.97366e-01   3.96206e-03
    2     0.30398070   5.66359e-07    -1.97081e-01   2.87374e-06
    3     0.30398357   2.94959e-13    -1.97081e-01   0.00000e+00
[...]
  V/F:           0.303984
  Component         z          x          y          K
  benzene         0.400000  0.333888  0.551373    1.6514
  toluene         0.600000  0.666112  0.448627    0.6735
[...]
PROBLEM SOLVED == PROBLEM POSED.  No divergence declared or imposed.
ASSUMPTIONS AND CAVEATS -- what this run announced along the way
  EXTRAPOLATIONS AND VALIDITY  (3)
    - component 'benzene': vapour pressure evaluated at T = 370 K, OUTSIDE its declared Trange (287.7
354.07) -- extrapolated, still returned
```

Listing 3: A refusal by name (the feed of Listing 1 with `vaporFraction 0.3;` added); exit status 2.

```
ERROR: stream state 'feed': OVER-SPECIFIED -- declares T = 370 K, P = 100000 Pa AND vaporFraction = 0.3
for a MIXTURE of 2 components (benzene, toluene).  A stream of known component flows is fixed by TWO
intensive variables (Duhem's theorem); [...] Remedies: (1) delete the `vaporFraction` line and let (T, P)
 resolve the split [...]; (2) if the QUALITY is the specification, the state is (P, q) or (T, q) and the
other variable must be SOLVED from it [...].
```

5.2. An integrated process with recycles: the Cavett problem

The Cavett problem [22] is a classical recycle benchmark: four flash drums at 800, 270, 49 and 13 psia, two mixers, three recycle streams (one liquid, two vapours) and a sixteen-component feed of 27 340.6 kmol/h entering the second stage (Figure 3). The case runs it on the specification of Rosen and Pauls [23] with SRK [24], every k_{ij} zero and declared, the mixers adiabatic, and Newton on the two mixer outlets as tear streams.

Table 4 keeps the three questions of Section 3.1 apart. The run converges in seven Newton iterations; Wegstein acceleration on the same problem limit-cycles and stops at its iteration limit, and the run exits with status 1 and writes no `converged/`. The converged plant closes its mass balance to 100.0000 %, every element to 0.0011 %, and its first law to +0.0019 kW of 396 325 kW exchanged; each of the six units leaves at most 0.0025 kW unattributed. This is the corrected case. In the first draft of this paper the same plant, converged, missed its first law by −549 319 kW (88 % of the energy exchanged), and the per-unit report put the whole gap on the two mixers: the adiabatic mixer published its outlet in the phase of its dominant inlet and computed its temperature on a different enthalpy surface from the one the report prices. The mixer now returns the state of equal enthalpy resolved at its own phase equilibrium, priced as the report prices it (the mixed streams flash, to 269.7 and 273.6 K), and both mixers close to 0.0000 kW. The earlier run is the example of the first two questions coming apart: numerically converged, physically inconsistent, and visibly so in the default report.

5.3. Nanofiltration of multi-ionic solutions

The engine models a spiral-wound or flat-cell nanofiltration element as a sequence of stations along the feed channel. At each station a transport law gives the ion fluxes

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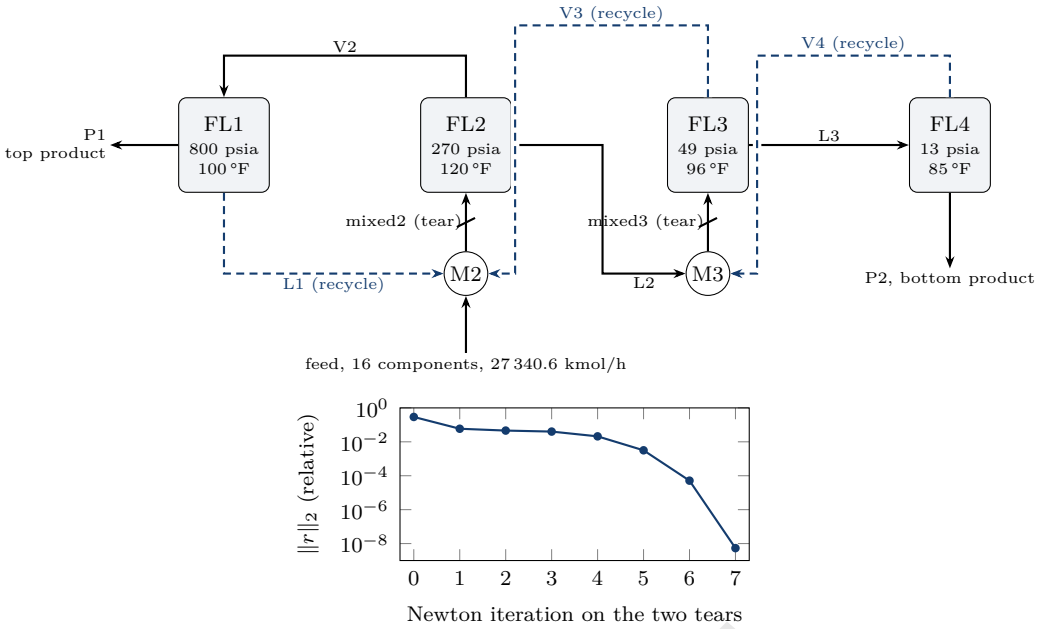


Figure 3: The Cavett problem as the case declares it (top): four isothermal flash drums, two adiabatic mixers, the two torn streams (slashed) and the three recycles (dashed). Bottom: the Newton history on the tears.

Table 4: The Cavett problem, the three questions of Section 3.1 answered separately (the run of this revision). Product flows are those of stream P1 (top product).

question	**evidence**	**verdict**
numerical convergence	Newton, 7 iterations, $\|r\|_2 = 5.4 \times 10^{-9}$ against 10^{-6}; Wegstein: 300 iterations, $\|\Delta\|_2 = 2.3 \times 10^{-2}$	converged (Newton); not converged, no answer written (Wegstein)
physical consistency	mass 100.0000 %; elements within 0.0011 %; first law +0.0019 kW of 396 325 kW exchanged	consistent
acceptance	P1: N_2 358.2, CH_4 2995.4, ethane 2383.8, isobutane 153.9 kmol/h; FLOWTRAN [23]: isobutane 198.2 kmol/h	agreement on the light ends; the loop intermediates differ with the K-model (SRK, $k_{ij} = 0$ here); no measurement exists

through the membrane, a polarisation model gives the concentrations at the membrane wall, and the water flux closes $J_v = A_w(\Delta P - \Delta\pi)$. Two transport laws are compared with measurements here.

Solution-diffusion-electromigration (SDEM). Following Yaroshchuk et al. [25], each ion i crosses the active layer (coordinate $\xi \in [0,1]$, potential ψ in units of RT/F) with a constant permeance P_i that already contains its partition coefficient:

$$j_i = -P_i\left(\frac{dc_i}{d\xi} + z_i c_i \frac{d\psi}{d\xi}\right), \qquad \sum_i z_i c_i = 0, \qquad \sum_i z_i j_i = 0, \qquad j_i = J_v\, c_{p,i}, \tag{5}$$

with c_i the virtual concentrations (of the solution that would be in equilibrium with each point of the layer), electroneutrality of that virtual solution, zero electric current and the permeate composition as the closure. The boundary conditions are the wall concentrations at $\xi = 0$ and the permeate at $\xi = 1$. The field follows from the last two conditions with no added parameter, $d\psi/d\xi = -\big(\sum_i z_i j_i/P_i\big)/\big(\sum_i z_i^2 c_i\big)$; for a single salt the system reduces exactly to $R = J_v/(J_v + P_s)$ with the ambipolar permeance

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$P_s = (z_+ + |z_-|)P_+P_-/(z_+P_+ + |z_-|P_-)$. The engine integrates Eq. (5) from the permeate face, where the unknown is, towards the wall, and solves for the permeate composition by Newton at each station.

Identifiability. A single-salt test determines one number, P_s, and not the two ion permeances: by the ambipolar relation, $1/P_s = (z_+/P_- + |z_-|/P_+)/(z_+ + |z_-|)$ is a linear combination of the reciprocal ion permeances. Several salts sharing ions give a linear system in those reciprocals; for the four salts of two cations and two anions its rank is four only because the valences differ (with equal valences one combination is lost). Ion permeances are therefore not calibrated from one salt; the source study determined them from multi-salt experiments with traces, and the engine reproduces the study's four salt permeances from its ion permeances to the table's rounding, which is a consistency check and not an identification.

Donnan–steric pore model (DSPM-DE). The second law resolves the pore: the extended Nernst–Planck flux of each ion with hindered convection, $N_i = -P_i(dc_i/d\xi + z_i c_i\, d\psi/d\xi) + K_{c,i}J_v c_i$, electroneutrality with the membrane's fixed charge X_d at every node, zero current, and steric, dielectric and Donnan partitions at both faces [26], with the hindrance factors of Dechadilok and Deen [27]. The pore is discretised on 101 nodes with exponentially fitted (Scharfetter–Gummel) fluxes [28] and every node, the wall and the water flux are solved together by Newton, as in the program NanoFiltran [29]. Against NanoFiltran, on 35 published problems, the two programs' grid-converged permeates agree within 7.9×10^{-7} relative. That is verification by comparison with another code; it says nothing about either code's agreement with a membrane.

Polarisation. The wall concentrations come from the multi-ionic film model of Geraldes and Afonso [30]: each ion has its own mass transfer coefficient from its own diffusivity, corrected for suction, and the ions are coupled by one interface potential fixed by electroneutrality at the wall, with no film thickness to declare. The data used below are *intrinsic* rejections (the authors corrected them for polarisation), so the cases make the film negligible ($k = 10\ \mathrm{m\,s^{-1}}$ in the fit, $k = 1\ \mathrm{m\,s^{-1}}$ in the trace probes) and compare intrinsic quantities.

Operating conditions and data. The measurements are those of Fernández de Labastida and Yaroshchuk [31] (CC BY 4.0): an NF270 membrane in a rotating-disk cell at 20 °C, four dominant salts (NaCl, $MgCl_2$, Na_2SO_4, $MgSO_4$) at 0.01 $\mathrm{mol\,L^{-1}}$ with $NaNO_3$ and NH_4Cl traces at 2×10^{-4} $\mathrm{mol\,L^{-1}}$, permeate fluxes 6–56 $\mu\mathrm{m\,s^{-1}}$, intrinsic rejections read from the paper's figures with the reading error recorded per point (the source publishes no measurement uncertainty).

Validation against measurements. Figure 4a is the DSPM-DE pore fitted to the measured rejections of the two chlorides: pore radius 0.675 ± 0.034 nm and fixed charge -10.5 ± 4.1 $\mathrm{mol\,m^{-3}}$ (95 % intervals; correlation -0.96), and the two sulfates, never seen by the fit, predicted with a mean absolute error of 0.93 rejection points against the 10 points the case declared before the fit. Two qualifications belong with that number. The partition into fit and held-out salts was chosen after two others had been measured and found not identifiable (the case says so), and held-out salts of the same study are not an independent study. And the sulfates' rejections are near unity, where a tight pore of almost any charge predicts them closely: on the transmission scale, $1 - R$, the prediction is 9–12 times too low for Na_2SO_4 and 49–67 times too low for $MgSO_4$. The acceptance criterion measured in

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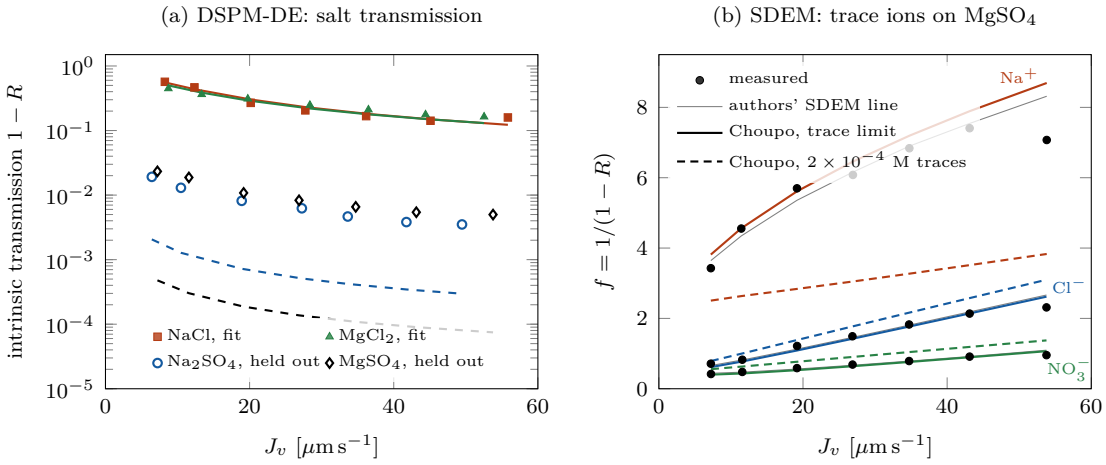


Figure 4: NF270 [31]: measurements (markers) against Choupo (lines). (a) DSPM-DE with its pore radius and fixed charge fitted to the two chlorides (solid lines) and the two sulfates predicted (dashed), on the transmission scale $1 - R$; on the rejection scale the sulfate points and lines overlap at $R \approx 0.98$–1. (b) SDEM with the paper's Table 1 permeances, trace ions on a 0.01 M $MgSO_4$ feed: Choupo in the trace limit the authors' analytical line assumes (solid), and at the experiments' own trace concentration (dashed); the thin grey line is the authors' own fit.

rejection points is passed; on transmission it would fail. The scale on which a membrane model is judged is part of the claim.

Figure 4b separates two things the first draft blurred. In the trace limit, which the authors' analytical line assumes, Choupo's SDEM reproduces that line within 3.5–5 % on average on the three trace ions shown, and the measurements within 7–8 %. At the experiments' own trace concentration it does not: the transmission of Na^+ rises by a factor of 1.5–2.3 and that of NO_3^- falls by a fifth to a third, and f departs from the measurements by 20–46 % on average. The reason is physical and is the source paper's own caveat: on this feed a trace nitrate at $P = 42\ \mu m\,s^{-1}$ carries more current than 0.01 M of sulfate at $0.13\ \mu m\,s^{-1}$, so the traces relax the field the trace-limit solution assumes they do not touch. The measured points follow the trace-limit line because the permeances were fitted *with* that approximation; the full model with the same permeances departs from the measurements. Parameters estimated under one approximation are not transferable to a model that drops it, and the consistent remedy, re-estimating the permeances with the full model, has not been done. Electrodialysis is treated with the same discipline: the limiting current of a bench stack, predicted from the ion diffusivities with no fitted transport number, lies +1 to +12 % (6.9 % on average) from the six measured values of Geraldes and Afonso [32].

5.4. A comparison with an open-source simulator

We compared Choupo with DWSIM [4], an open-source (GPL-3.0) simulator, on five samples from the library it ships. Each sample was rewritten as a Choupo case from its published problem; no file or property parameter of the sample was copied, because the samples embed pair parameters from databanks this project excludes. The reference numbers are the solved states stored in the sample files, read at one commit of the reference's repository; the samples were not re-executed in an identified version of the reference, so the comparison is with stored states, and an inconsistency found in a stored state is a fact about the file, not necessarily about the program. Following the review of

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Table 5: Case E: Choupo's methane conversion against the number of RK4 steps through the bed, with the article's rate laws and with the sample's stored ones. The sample stores 0.7532.

steps	article rate laws	sample's rate laws
400	0.7490	0.7510
1 600	0.7516	0.7539
6 400	0.7549	0.7573
25 600	0.7554	0.7578
102 400	0.7554	0.7578

the first draft, the cases are split into two exercises, and the audit of the stored files is in the supplement (S2).

Exercise 1: the same equations. Only case E, a methane steam reformer, could be posed with identical equations on both sides. One isothermal plug-flow reactor (1 m^3, 1000 K, 1 atm, 0.386 kg of catalyst per m^3 as the sample has it) runs the three rates of Xu and Froment [33]. In a variant of the case, the sample's own stored rate expressions are written into Choupo exactly as stored – partial pressures in atm, its rounded pre-exponential factors, its own equilibrium-constant correlations and its two truncated adsorption denominators – so that the two programs integrate the same rate laws over the same volume, flow and catalyst basis. The equation of state is immaterial at 1 atm (setting every k_{ij} to zero moves no printed digit). Choupo's integration was first checked for grid convergence, and the check changed the case: the inlet carries 1 % hydrogen and every rate divides by a power of its partial pressure, so the first millimetres are stiff, and the 400 steps the case first used left the methane conversion 0.85 % short of its limit – more than the gap it had been compared on (Table 5). At 25 600 steps, with the same rate laws, Choupo's conversion is 0.7578 against the stored 0.7532 (+0.61 %), CO_2 −3.0 % and water −1.7 %. What remains is the implementation: the integrators, and an Ergun pressure drop of 584 Pa the case does not reproduce, which was measured not to explain it (applied as a mean pressure it moves the conversion to 0.7581, the other way). The sample stores a volume increment of 0.05 m^3, twenty increments; coarse integration lowers Choupo's conversion, which is consistent with the stored value being lower, but the stored state does not say how each increment is integrated and that remains a hypothesis. The duty does not depend on the rate laws: 466.4 kW in Choupo, the heat of reaction of its own conversion, against a stored 1221.8 kW that is 2.6 times the heat of reaction of the conversion the sample stores.

Exercise 2: different parameters and different models. In the other four cases the two programs solve the same flowsheet with different property data, and in some with different models; the differences are then part of what is studied, and each conclusion is limited to what the runs show (Table 6).

A Cavett's train as the sample builds it, with compressors and letdown valves. Both programs use the Peng–Robinson equation [34]; Choupo with its catalogue's critical constants and group-contribution k_{ij} evaluated at each separator's temperature, the reference with its own databank. Products agree to 0.5 % and compressor outlets to 0.35 K; the valve outlets are 1.5–7.9 K warmer in Choupo. Letting the sample's own stored liquid down through Choupo's valve gives 288.61 K against the stored 281.65 K, and 293.22 K with every k_{ij} zero: the outlet is sensitive to the thermodynamic parameters by several kelvin. This shows sensitivity; it does not exclude a difference in the valve implementations, which the comparison cannot separate without the reference's parameters. With adiabatic mixers, as the sample has them, Choupo's plant

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Table 6: The five benchmark cases: what each compares, and the headline difference (Choupo against the reference's stored state).

case	what is equal, what differs	headline
E. steam reformer (exercise 1)	same rate laws (the sample's own), same reactor; integrators differ	CH_4 conversion 0.7578 vs 0.7532 (+0.61 %); duty 466.4 vs 1221.8 kW
A. Cavett with compressors	same equation of state; critical constants and k_{ij} differ	products ±0.5 %; valve outlets +1.5 to +7.9 K
B. pressure-swing distillation	same flowsheet; NRTL pair and UNIFAC vs the reference's own parameters	duties −0.7 to +1.5 %; distillate x −0.017, −0.022
D. methanol from CO_2	same plant up to the make-up; a knock-out drum downstream; different kinetic constants	CO_2 compressors within 5 %; product 1.3967 vs 1.2304 mol s^{-1}
C. extraction with MIBK	adiabatic vs isothermal: different problems	supplement only

first law closes to −69.1 kW and each mixer to 0.0000 kW (with the mixers at declared temperatures, −67.3 kW over 54.6 MW exchanged, attributed by the engine's report to declared model-boundary steps and the compressors' outlet-temperature tolerance).

B Methanol–acetone pressure-swing distillation, 1 and 6 atm. Choupo ran it with its own NRTL pair (Section 4.3; not validated, and 50 K beyond its evidence at 6 atm, which every run announces) and with UNIFAC. The two Choupo variants agree with each other to 0.006 in every distillate composition and 1 kW in every duty; the four duties are within −0.7 to +1.5 % of the reference and the distillate compositions 0.017 and 0.022 lower. The reference's stored flows do not close by more than those differences (+0.14 % on the plant), so the comparison cannot resolve them. The acetone product leaves pure at 6 atm 2.05 K cooler in Choupo, a difference between the two vapour-pressure correlations. Choupo specifies distillate rates where the sample specifies bottoms flows that admit no steady state.

D Methanol from CO_2, the whole plant, its reactor on the kinetics of Vanden Bussche and Froment [35] as the article prints them (the sample's depart from the article in four places, listed in supplement S2). The hydrogen feed is stored two-phase and a Choupo compressor refuses liquid in its suction, so a knock-out drum, not in the sample, removes 0.56 mol s^{-1} of water before the hydrogen compressor. The two plants therefore have different feeds downstream of the make-up, and only the CO_2 compression train is like for like (compressors within 0.02–4.9 %). Downstream the methanol product is 1.3967 against 1.2304 mol s^{-1} and the hydrogen compressor 140.00 against 129.74 kW; most of that gap is the departure, not a model difference. Choupo's plant closes its first law to 0.62 kW. The second shortcut column's minimum stage number differs by a factor of two (9.21 against 4.55); at the products' own recoveries the reference's value corresponds to a mean methanol/water relative volatility of 7.35, where one measured point at $x_{\mathrm{methanol}} = 0.969$ gives 2.43 – a single composition, not the column average Fenske's equation needs, so we draw no conclusion on which is right.

Case C (acetone extraction with MIBK) is not compared here: the reference column is adiabatic and cools by 13.4 K while Choupo's extractor is isothermal, and temperature moves the liquid–liquid split, so the two programs solve different problems. It is in the supplement, with its curated liquid–liquid set.

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Table 7: Defects in Choupo that the comparison exposed, and their status on the tree of this revision. "Silent" means the run published a result at exit zero.

case	defect	now
B	a multi-feed column accepted a feed on its top stage and left it out of the balance: 109 % mass closure (silent)	refused by name
B	the total condenser published one more equilibrium contact than a total condenser	condenser type declared; stage count is equilibrium stages
C	the liquid–liquid flash returned one liquid on feeds its own stability test found unstable; the stability test omitted $\ln\gamma$ (silent)	fixed; a split is finished by Newton on equal activities
C	the extractor published a non-converged cascade with 0.229 mass closure (silent)	refused, with the closure and the stages
D	three numerical defects in the phase changer, the regression identifiability and the bubble-pressure search (two silent)	fixed
A	the adiabatic mixer created energy (9 685 and 40 204 kW)	fixed (Section 5.2)
D	the compressor treated a two-phase feed as vapour, in silence	refuses liquid in its suction
D	the shortcut column publishes both products at the feed's bubble point; the PFR's printed summary gives the outlet flow 1000 times too small (its published indicator is right)	named, not fixed

What the comparison found in Choupo. The comparison was run to find defects, and Table 7 lists those it found on Choupo's side. Five were silent – a result published at exit zero that the engine should have refused or computed otherwise – and each now refuses by name or is fixed; the corrections that moved reference answers of the corpus were listed row by row before anything was re-recorded. Two named defects remain open. The benchmark cases are not part of the regression suite: a check that pinned their numbers would pin the reference's stored inconsistencies too. The reviewer's point stands that Choupo's own side can be tested without agreeing with the reference, and what each case asserts about Choupo alone – that it runs, converges, carries no NaN and closes its own first law, and that a case-local copy of a catalogue record is still a copy – is checked by the comparison script on every run; turning that into a check of the regression suite is planned.

6. Discussion and limitations

Choupo does not aim at the breadth of industrial simulators: its catalogue is curated rather than encyclopaedic, it has no equation-oriented solver, and its models cover what its tutorials exercise. The limitations we know of are these.

- *Validation is a small subset.* No rigorous column, recycle plant or liquid–liquid flowsheet is anchored to a measurement; the membrane fit of Section 5.3 is validated within one study, on a scale that flatters it.
- *Open first-law defects are listed, not closed.* Twelve steady plants do not close their first law within the threshold; they are listed by name with their residuals.
- *Some balances are not measured.* The first law of the time-integrated batch reactor and dynamic CSTR is claimed neither way.
- *A check's reach is its scope*, and the benchmark cases are outside the regression suite.
- *Curated data are mostly unreviewed*: the regressed pairs' measurements were transcribed from the archive and have not been read back against the articles, and four records with an encumbered source await replacement.
- *The SDEM permeances* were estimated by their authors under the trace approximation and are used here as published; re-estimating them with the full model is open.
- *The pedagogical value is untested.*

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7. Conclusions

Choupo enforces, where every input is read, that the data, hypotheses, equations, parameter evidence, iterations and balances of a calculation are observable in files and in the log, and it verifies the consequence across a corpus. The verification method keeps three questions apart: the Cavett problem converged in both drafts of this paper, but was physically consistent only in the second, and the default balance report is what showed the difference. The curation rule accepts a regressed parameter set only against an independent study and a band declared before the fit; the band is conservative for a correct model, and the example verdict survives a χ^2 test with every point's own uncertainty. In nanofiltration, a fitted pore predicts held-out salts in rejection points but not in transmission, and published ion permeances reproduce their own trace-limit fit but not the measurements at the experiments' trace concentration. With identical rate laws, two simulators still differ by 0.6 % in conversion, after one of them found its own grid unconverged. What remains is stated as plainly: the validation subset is small, open first-law defects are listed rather than closed, the curated pairs are not read back against their articles, and the pedagogical value has not been measured.

CRediT authorship contribution statement

Vítor Geraldes: Conceptualization, Software (architecture), Supervision, Writing – review & editing. **Pedro Mendes:** TODO. **Miguel Rodrigues:** TODO.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. For transparency: the name *Choupo* is a registered trade mark of TalentGround Lda., a holding company of Vítor Geraldes's family. The software is free under the GNU General Public License, version 3 or later; the trade mark is not licensed by it, and no commercial licence of the software is offered.

Data and code availability

Choupo's source, tutorials, guides and browser application are public under the GNU General Public License, version 3 or later, at `https://github.com/choupo-admin/choupo`; the browser application runs at `https://www.choupo.org`. *The version evaluated in this paper* is the development line at the commit that carries this manuscript (branch `claude/c126-papers-revision`, based on `c11d0354a`); every listing, figure and table was regenerated on it, and the supplement (S1) gives, for each, the case, the command and the output file it was read from. A release tag cut from that commit and an archived copy with a DOI: TODO (B4, B6). The benchmark cases are under `benchmarks/dwsim/`, each with a README; `benchmarks/dwsim/compare.py` runs them.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work the authors used Claude Code (Anthropic), an agentic coding assistant running models of the Claude Opus and Claude Fable families, in order to implement substantial parts of the software described here, its documentation

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and its tutorial corpus, and to draft and revise this manuscript from the project's records, its runs and the authors' written briefs. Choupo was conceived and architected by Vítor Geraldes; the agents work under written briefs, and architecture, pedagogical scope and a recorded list of reserved decisions remain with him. The published project is curated, reviewed, corrected and maintained by Vítor Geraldes and Pedro Mendes. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Acknowledgements

Funding and institutional support: TODO. OPENFOAM® is a registered trade mark of OpenCFD Limited, producer and distributor of the OpenFOAM software via www.openfoam.com; Choupo is an independent project, not affiliated with OpenCFD Ltd or the OpenFOAM Foundation. DWSIM is named only as the open-source software compared in Section 5.4; no endorsement by its authors is implied. The nanofiltration measurements are those of Fernández de Labastida and Yaroshchuk [31], published under CC BY 4.0.

Appendix A. Automated checks behind the examples

The repository's generated manifest lists 257 automated checks, each with the sentence its own output states as its claim, including what it does not check. Table A.1 lists the ones named in this paper.

Table A.1: The automated checks named in this paper, and the class of error each keeps from passing in silence on the cases it runs.

The class of error	Check
a plant that does not conserve mass (engine's own report, every steady case)	`check_mass_closure`
a plant whose atoms do not close	`check_element_balance`
a first law that does not close, or a listed one that moves	`check_energy_closure`
a unit computing with a state its streams do not carry	`check_feed_thermal_state`
a stream declared by three variables where two fix it	`check_overspecified_stream`
a model word the engine does not know, taken as the default	`check_model_default_registered`
an approximation nobody authorised	`check_problem_divergence`
a promoted pair that drifts from its evidence, or splits the liquid	`check_regressed_pairs`
an SDEM permeate that is not electroneutral, or that drifts from the digitised reference lines	`check_sdem`
a pore solver that drifts from NanoFiltran	`check_nanofiltran_reference`
a fitted membrane record used after its data changed	`check_fit_then_simulate`
a value whose licence does not allow redistribution	`check_source_licence`
a validation case that joins or leaves the subset unnoticed	`check_validation_subset`

References

[1] A. Lee, J. H. Ghouse, J. C. Eslick, C. D. Laird, J. D. Siirola, M. A. Zamarripa, D. Gunter, J. H. Shinn, A. W. Dowling, D. Bhattacharyya, L. T. Biegler, A. P. Burgard, D. C. Miller, The IDAES process modeling framework and model library – flexibility for process simulation and optimization, Journal of Advanced Manufacturing and Processing 3 (2021) e10095. doi:10.1002/amp2.10095.

DRAFT

[2] I. H. Bell, J. Wronski, S. Quoilin, V. Lemort, Pure and pseudo-pure fluid thermophysical property evaluation and the open-source thermophysical property library CoolProp, Industrial & Engineering Chemistry Research 53 (2014) 2498–2508. doi:10.1021/ie4033999.

[3] D. G. Goodwin, H. K. Moffat, I. Schoegl, R. L. Speth, B. W. Weber, Cantera: An object-oriented software toolkit for chemical kinetics, thermodynamics, and transport processes, 2025. URL: https://www.cantera.org. doi:10.5281/zenodo.17620923, tODO (authors): the Zenodo DOI is a DataCite DOI and could not be checked through Crossref; confirm it on zenodo.org.

[4] D. W. O. d. Medeiros, DWSIM: an open-source chemical process simulator, https://github.com/DanWBR/dwsim, 2024. GPL-3.0 (README and COPYING read at commit 1abf72d1b6b41d3e9a8cc770d3cc4e8fc76e5766, 2026-07-17); sample files carrying BuildVersion 8.8.0.0.

[5] J. H. Wegstein, Accelerating convergence of iterative processes, Communications of the ACM 1 (1958) 9–13. doi:10.1145/368861.368871.

[6] J. A. Nelder, R. Mead, A simplex method for function minimization, The Computer Journal 7 (1965) 308–313. doi:10.1093/comjnl/7.4.308.

[7] D. W. Marquardt, An algorithm for least-squares estimation of nonlinear parameters, Journal of the Society for Industrial and Applied Mathematics 11 (1963) 431–441. doi:10.1137/0111030.

[8] M. L. Michelsen, The isothermal flash problem. part i. stability, Fluid Phase Equilibria 9 (1982) 1–19. doi:10.1016/0378-3812(82)85001-2.

[9] H. G. Weller, G. Tabor, H. Jasak, C. Fureby, A tensorial approach to computational continuum mechanics using object-oriented techniques, Computers in Physics 12 (1998) 620–631. doi:10.1063/1.168744.

[10] W. L. Oberkampf, C. J. Roy, Verification and Validation in Scientific Computing, Cambridge University Press, Cambridge, 2010. doi:10.1017/CBO9780511760396.

[11] National Aeronautics and Space Administration, Standard for Models and Simulations, Technical Report NASA-STD-7009, NASA, 2008. TODO (authors): cite the revision read (7009 of 2008, or 7009A of 2016).

[12] T. J. Williams, R. E. Otto, A generalized chemical processing model for the investigation of computer control, Transactions of the American Institute of Electrical Engineers, Part I: Communication and Electronics 79 (1960) 458–473. doi:10.1109/TCE.1960.6367296.

[13] W. Wagner, J. R. Cooper, A. Dittmann, J. Kijima, H.-J. Kretzschmar, A. Kruse, R. Mareš, K. Oguchi, H. Sato, I. Stöcker, O. Šifner, Y. Takaishi, I. Tanishita, J. Trübenbach, T. Willkommen, The IAPWS industrial formulation 1997 for the thermodynamic properties of water and steam, Journal of Engineering for Gas Turbines and Power 122 (2000) 150–184. doi:10.1115/1.483186.

DRAFT

[14] W. J. Hamer, Y.-C. Wu, Osmotic coefficients and mean activity coefficients of uni-univalent electrolytes in water at 25 °C, Journal of Physical and Chemical Reference Data 1 (1972) 1047–1100. doi:10.1063/1.3253108.

[15] L. N. Plummer, E. Busenberg, The solubilities of calcite, aragonite and vaterite in CO_2–H_2O solutions between 0 and 90 °C, and an evaluation of the aqueous model for the system $CaCO_3$–CO_2–H_2O, Geochimica et Cosmochimica Acta 46 (1982) 1011–1040. doi:10.1016/0016-7037(82)90056-4.

[16] Joint Committee for Guides in Metrology, Evaluation of measurement data – Guide to the expression of uncertainty in measurement, Technical Report JCGM 100:2008, JCGM, 2008.

[17] M. Frenkel, R. D. Chirico, V. V. Diky, Q. Dong, S. Frenkel, P. R. Franchois, D. L. Embry, T. L. Teague, K. N. Marsh, R. C. Wilhoit, ThermoML – an XML-based approach for storage and exchange of experimental and critically evaluated thermophysical and thermochemical property data. 1. experimental data, Journal of Chemical & Engineering Data 48 (2003) 2–13. doi:10.1021/je025645o.

[18] X. Chen, B. Yang, A. A. Abdeltawab, S. Al-Deyab, G. Yu, X. Yong, Isobaric vapor–liquid equilibrium for acetone + methanol + phosphate ionic liquids, Journal of Chemical & Engineering Data 60 (2015) 612–620. doi:10.1021/je5007373.

[19] A. V. Orchillés, P. J. Miguel, E. Vercher, A. Martínez-Andreu, Ionic liquids as entrainers in extractive distillation: Isobaric vapor–liquid equilibria for acetone + methanol + 1-ethyl-3-methylimidazolium trifluoromethanesulfonate, Journal of Chemical & Engineering Data 52 (2007) 141–147. doi:10.1021/je0603170.

[20] A. Fredenslund, R. L. Jones, J. M. Prausnitz, Group-contribution estimation of activity coefficients in nonideal liquid mixtures, AIChE Journal 21 (1975) 1086–1099. doi:10.1002/aic.690210607.

[21] H. Renon, J. M. Prausnitz, Local compositions in thermodynamic excess functions for liquid mixtures, AIChE Journal 14 (1968) 135–144. doi:10.1002/aic.690140124.

[22] R. H. Cavett, Application of numerical methods to the convergence of simulated processes involving recycle loops, in: Proceedings of the American Petroleum Institute, volume 43, 1963, p. 57. TODO (authors): volume and page as the literature cites them; not checked against the proceedings (no DOI).

[23] E. M. Rosen, A. C. Pauls, Computer aided chemical process design: the FLOWTRAN system, Computers & Chemical Engineering 1 (1977) 11–21. doi:10.1016/0098-1354(77)80003-3.

[24] G. Soave, Equilibrium constants from a modified Redlich-Kwong equation of state, Chemical Engineering Science 27 (1972) 1197–1203. doi:10.1016/0009-2509(72)80096-4.

[25] A. Yaroshchuk, M. L. Bruening, E. E. Licón Bernal, Solution-diffusion–electromigration model and its uses for analysis of nanofiltration, pressure-retarded osmosis and forward osmosis in multi-ionic solutions, Journal of Membrane Science 447 (2013) 463–476. doi:10.1016/j.memsci.2013.07.047.

DRAFT

[26] W. R. Bowen, J. S. Welfoot, Modelling the performance of membrane nanofiltration – critical assessment and model development, Chemical Engineering Science 57 (2002) 1121–1137. doi:10.1016/S0009-2509(01)00413-4.

[27] P. Dechadilok, W. M. Deen, Hindrance factors for diffusion and convection in pores, Industrial & Engineering Chemistry Research 45 (2006) 6953–6959. doi:10.1021/ie051387n.

[28] D. L. Scharfetter, H. K. Gummel, Large-signal analysis of a silicon Read diode oscillator, IEEE Transactions on Electron Devices 16 (1969) 64–77. doi:10.1109/T-ED.1969.16566.

[29] V. Geraldes, A. M. Brites Alves, Computer program for simulation of mass transport in nanofiltration membranes, Journal of Membrane Science 321 (2008) 172–182. doi:10.1016/j.memsci.2008.04.054.

[30] V. Geraldes, M. D. Afonso, Prediction of the concentration polarization in the nanofiltration/reverse osmosis of dilute multi-ionic solutions, Journal of Membrane Science 300 (2007) 20–27. doi:10.1016/j.memsci.2007.04.025.

[31] M. Fernández de Labastida, A. Yaroshchuk, Nanofiltration of multi-ion solutions: Quantitative control of concentration polarization and interpretation by solution-diffusion-electro-migration model, Membranes 11 (2021) 272. doi:10.3390/membranes11040272, cC BY 4.0.

[32] V. Geraldes, M. D. Afonso, Limiting current density in the electrodialysis of multi-ionic solutions, Journal of Membrane Science 360 (2010) 499–508. doi:10.1016/j.memsci.2010.05.054.

[33] J. Xu, G. F. Froment, Methane steam reforming, methanation and water-gas shift: I. intrinsic kinetics, AIChE Journal 35 (1989) 88–96. doi:10.1002/aic.690350109.

[34] D.-Y. Peng, D. B. Robinson, A new two-constant equation of state, Industrial & Engineering Chemistry Fundamentals 15 (1976) 59–64. doi:10.1021/i160057a011.

[35] K. M. Vanden Bussche, G. F. Froment, A steady-state kinetic model for methanol synthesis and the water gas shift reaction on a commercial $Cu/ZnO/Al_2O_3$ catalyst, Journal of Catalysis 161 (1996) 1–10. doi:10.1006/jcat.1996.0156.

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